Crack the Data Analytics Interview in 7 Days

A Complete Handbook for Sri Vaishnavi Devarashetty

About This Handbook

This handbook is your complete playbook for landing an Analytics Manager, Senior Analytics Lead, or AI-Augmented Data Analyst role. Every section is built from your actual 12+ years of experience at NVIDIA and Apple (via Wipro) --- not generic advice. Use it as your daily companion across the 7 days.

Your strongest assets going in:

- 12+ years of progressive analytics leadership
- NVIDIA brand + Apple (Wipro client) on the same resume
- Real Tableau, Snowflake, and SQL depth --- defensible in any interview
- Already mentored and promoted analysts
- Hands-on AI tool usage with a governance discipline that separates you from the crowd

Your live portfolio: sri-vaishnavi-1988.github.io

Your skills showcase: sri-vaishnavi-1988.github.io/skills.html

Your interview quiz: sri-vaishnavi-1988.github.io/quiz-app/

Part 1: Your Positioning

The Headline

> Senior Analytics Leader · 12+ yrs Tableau, Snowflake, SQL · Building AI-assisted analytics workflows that turn dashboards into decisions

Use this verbatim on LinkedIn and at the top of every interview introduction.

The 30-Second Verbal Pitch

> "I am a senior analytics professional with 12+ years across NVIDIA and Apple, where I worked as a Wipro client engagement lead. I am deep in Tableau, Snowflake, SQL, and Python. Over the last year I have been integrating AI tools like ChatGPT, GitHub Copilot, and Claude into my day-to-day work to accelerate SQL writing, dashboard documentation, and stakeholder reporting. I am now exploring agentic AI on platforms like Snowflake Cortex. My value is not just using AI --- it is translating business questions into analytics solutions that AI can help scale."

Deliver this in the first two minutes of every interview. It frames everything that follows.

Why This Positioning Works

- Names real enterprise brands for instant credibility
- Lists only tools you actually use --- fully defensible under questioning
- Distinguishes "uses AI" (commodity) from "translates business questions into AI-scalable analytics" (leadership signal)
- Targets the Analytics Manager and Lead Analyst bracket where Indian hiring is strong

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Part 2: The 7-Day Daily Plan

Day 1: Foundation and Mindset

Morning (2 hours)

Start here. Your goal today is clarity --- not cramming.

1. Read this entire handbook once through without stopping
2. Open your LinkedIn and update your headline to the exact version in Part 1
3. Update your resume summary to match your verbal pitch
4. Open your GitHub portfolio and confirm all links work: portfolio, skills showcase, and quiz

    Afternoon (2 hours)

    Answer these questions in writing. The act of writing them forces you to think:

    - What is the single most impactful dashboard I built? What did it change for the business?
    - What is one time I caught a major data quality issue before it reached stakeholders?
    - What is one time I used AI to speed up my work --- and what human review did I add?
    - What does a junior analyst struggle with that I no longer struggle with?

    These become the raw material for your STAR stories in Day 3.

    Evening (30 minutes)

    Take the interview quiz on your portfolio at sri-vaishnavi-1988.github.io/quiz-app/ in Study Mode. Do not worry about your score. Just identify which categories feel weakest.

Day 2: SQL and Snowflake Deep Dive

SQL is tested in almost every analytics interview. Expect at least one live coding question.

Morning: Window Functions (1.5 hours)

Open `sql/01_window_functions.sql` from your portfolio. Read each query. Then close the file and rewrite each one from memory. The five patterns to own:

Running Total ```sql SUM(completion_count) OVER (PARTITION BY learner_id ORDER BY completion_date ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS running_total ```

Month-over-Month Change ```sql LAG(completions) OVER (ORDER BY month) AS prev_month, completions - LAG(completions) OVER (ORDER BY month) AS mom_change ```

Top N Per Group (with QUALIFY) ```sql RANK() OVER (PARTITION BY department ORDER BY score DESC) AS rnk QUALIFY rnk <= 3 ```

Percentile Ranking ```sql PERCENT_RANK() OVER (PARTITION BY course ORDER BY score) AS pct_rank ```

Days Since Last Activity ```sql DATEDIFF('day', last_activity_date, CURRENT_DATE) AS days_inactive ```

Afternoon: Snowflake Optimization (1.5 hours)

Open `sql/02_snowflake_optimization.sql`. These are the patterns interviewers ask about when they want to know if you are a real Snowflake practitioner:

- **Clustering keys** --- when and why to use them on fact tables
- **QUALIFY** --- why it is faster than a wrapping subquery
- **Zero-copy CLONE** --- how to create test environments without storage cost
- **Time Travel** --- how to detect pipeline data drops
- **Result cache** --- when to disable it for live KPI spot checks

Interview Answer: "How do you optimize a slow Snowflake query?"

> "I start by checking if the query is scanning too many micro-partitions --- usually because filters are not on the clustering key. Then I look for non-sargable WHERE clauses like functions wrapped around indexed columns. For complex queries I use CTEs to avoid repeated subquery evaluation and use QUALIFY instead of outer SELECT for window function filters. If the data is large and frequently filtered on date ranges, I add a clustering key on the date column."

Day 3: STAR Stories --- Your Experience Made Memorable

Every behavioral question has the same structure: Situation, Task, Action, Result. Interviewers remember stories, not bullet points.

Build These 5 Stories Today

Story 1: The Dashboard That Changed Something

Situation: Apple L&D needed visibility into course completion across global teams. Data was fragmented across multiple learning platforms.

Task: Build a unified Tableau dashboard that program owners could use to make decisions about content and delivery.

Action: Integrated multiple L&D data sources into Snowflake, designed a dimensional model, built calculated fields and LOD expressions, applied row-level security for department visibility.

Result: User satisfaction improved 20%. Content delivery costs reduced 10%. Processing time cut 30%.

Story 2: The Data Quality Issue You Caught

Situation: A routine pipeline refresh at Apple produced completion rates that were 15% higher than the previous week with no program change.

Task: Investigate before the weekly stakeholder briefing goes out.

Action: Used Snowflake Time Travel to compare current and previous day row counts. Found a duplicate load from a source system. Flagged to the data engineering team. Held the briefing.

Result: Prevented incorrect data from reaching Apple program owners. Established a daily row count reconciliation check as a standard governance step.

Story 3: Leading Without Authority

Situation: At NVIDIA, three junior analysts were building dashboards with inconsistent KPI definitions. Two different definitions of "active employee" were in production.

Task: Standardise without having formal management authority over the team.

Action: Convened a KPI alignment session. Documented agreed definitions in a shared data dictionary. Built a reusable Snowflake view as the single source of truth. Ran SQL and Tableau training sessions.

Result: KPI consistency across 5+ analytics products. Two of the three analysts were promoted within 18 months.

Story 4: AI Saves the Day (With Governance)

Situation: A stakeholder at Apple requested a complex Snowflake query to analyse learner drop-off across 40+ courses in 12 regions with multi-level filtering.

Task: Deliver the analysis within two hours.

Action: Used GitHub Copilot to scaffold the CTE structure and window functions. Reviewed every generated line before running. Caught one incorrect PARTITION BY that would have inflated regional figures. Validated output against a manual spot check on three departments.

Result: Delivered in 90 minutes. Accuracy confirmed. The governance step --- reviewing before running --- caught the error.

Story 5: Mentoring That Compounded

Situation: A junior analyst at NVIDIA was building reports in Excel that took four hours each week. She had no SQL experience.

Task: Upskill her so the team could free her time for higher-value work.

Action: Ran four one-hour SQL training sessions. Built template queries she could adapt. Reviewed her first five Snowflake queries before she ran them in production.

Result: Her weekly report dropped from four hours to 20 minutes. She built two Tableau dashboards independently within three months. She was promoted within 18 months.

Day 4: Tableau Mastery

Tableau is your deepest skill. Use Day 4 to make sure you can talk about it at a level that signals genuine expertise, not just user familiarity.

LOD Expressions --- The Interview Separator

Most analysts have used Tableau. Few can explain LOD expressions clearly. This is where you separate yourself.

FIXED --- The Most Common

Computes at a specified dimension regardless of what is in the view.

``` { FIXED [Department] : AVG([Completion Rate]) } ```

Use case: Show each course row alongside its department average. Even when the view is filtered to one course category, the department average stays stable.

Interview explanation: "FIXED ignores dimension filters in the view. It only respects data source filters and context filters. I use it when I need a benchmark that should not change when the user drills down."

INCLUDE --- Add Granularity

``` { INCLUDE [Course Name] : AVG([Score]) } ```

Use case: When the view is at department level, force the average to consider individual course scores to avoid Simpson's paradox.

EXCLUDE --- Percent of Total

``` SUM([Completions]) / { EXCLUDE [Course Name] : SUM([Completions]) } ```

Use case: Each bar shows its share of total without needing a table calculation.

Context Filters --- The Hidden Gotcha

Interviewers often ask why a Top N filter is not working. The answer is almost always context filters.

> "Context filters execute first and create a temporary dataset. All other filters --- including Top N --- operate on that reduced set. Without a context filter, a Top N across a dimension that is also filtered can produce unexpected results because Top N sees the full dataset before the dimension filter applies."

Performance Optimisation

When asked how you keep Tableau dashboards fast:

1. Pre-aggregate in Snowflake SQL --- push computation to the warehouse
2. Use extract instead of live connection for large datasets
3. Avoid LOD expressions on very high cardinality fields
4. Use context filters correctly so other filters operate on smaller sets
5. Limit the number of marks in a view --- more marks means more rendering time
```

Day 5: Python and Data Governance

Python --- The Practical Interview

You are unlikely to be asked to build a machine learning model. You will be asked about data cleaning, pipeline design, and automation. Focus here.

The 5 pandas Operations Every Interviewer Asks About

```python

# 1. Read and basic shape

df = pd.read_csv("data.csv", parse_dates=["enrollment_date"]) df.shape, df.dtypes, df.isnull().sum()
```

2. Handle missing values (never use 0 for numeric missing)

```
df["score"] = df["score"].where(df["status"] == "completed", other=np.nan)
```

3. Remove duplicates on specific columns

```
df = df.drop_duplicates(subset=["learner_id", "course_id"], keep="last")
```

4. GroupBy with multiple aggregations

summary = df.groupby("department").agg(total_enrollments=("learner_id", "count"), avg_score=("score", "mean"), completion_rate=("status", lambda x: (x == "completed").mean())).reset_index()

5. Apply a custom function

df["risk_flag"] = df.apply(lambda row: "High" if row["days_inactive"] > 30 and row["pct_complete"] < 20 else "Medium" if row["days_inactive"] > 14 and row["pct_complete"] < 50 else "Low", axis=1) ```

The 80/20 Rule Question

Every interviewer asks some version of: "What do you spend most of your time on?"

> "About 80% of project time goes into data preparation --- cleaning, validation, joining, and ensuring the pipeline is reliable. Only 20% is the actual analysis or visualisation. The AI tools I use, particularly GitHub Copilot for SQL and Python scaffolding, have started to compress that 80% meaningfully. But the human review step still sits with me."

Data Governance --- The Leadership Signal

Junior analysts talk about governance as a compliance burden. Managers talk about it as a risk and trust function. Frame yours as the latter.

Your Pre-Publish Checklist (from real practice at Apple):

1. Row count matches expected range versus yesterday
2. No NULLs in primary key columns
3. Date range covers the full reporting period
4. KPI totals cross-check against source system
5. Tableau filters apply correctly to all sheets
6. LOD expressions validated against raw SQL
7. PII fields masked or excluded
8. Stakeholder sign-off received before go-live

   When asked "How do you ensure data quality?"

   > "I run a pre-publish checklist before anything goes to stakeholders. It covers source validation, KPI cross-checks, Tableau filter behaviour, and governance compliance. At Apple, this discipline reduced compliance-related issues by 20%. I also use Snowflake Time Travel for pipeline reconciliation --- comparing current row counts to yesterday to catch silent drops before anyone reports a number."

   ## Day 6: AI Fluency and Behavioural Questions

The AI Question --- You Will Be Asked This

Interviewers are currently sorting candidates into three buckets:

1. "Has not used AI at all" --- falling behind
2. "Pastes AI output into reports unchecked" --- dangerous
3. "Uses AI with a governance discipline" --- the hire

You are bucket three. Make that clear.

The Framework Answer

When asked "How do you use AI in your day-to-day?"

1. **The tool.** "I use ChatGPT, GitHub Copilot, and Claude."
2. **The use case.** "Primarily SQL acceleration, dashboard documentation, and translating analyst findings into business language."
3. **The governance.** "Every AI output is reviewed before it leaves my hands --- numbers, queries, and stakeholder messages. That review step is non-negotiable."
4. **The next step.** "I am now exploring Snowflake Cortex and agentic workflows for routine reporting automation."

What Is Agentic AI?

> "Agentic AI is when an AI system can plan and take multi-step actions autonomously --- pulling data, running a query, and posting a summary without a human prompt at each step. In analytics, it means moving from dashboards that wait for users to systems that actively surface insights when they matter. I have used agentic tools and am exploring frameworks like Snowflake Cortex for this. I have not deployed one in production yet, and I would want to do that with proper guardrails in place."

The honesty in that last sentence is a leadership signal. Inflated AI claims get caught in two follow-up questions.

Behavioural Questions and Your Answers

"Tell me about a time you influenced without authority."

Use Story 3 from Day 3 --- the KPI standardisation at NVIDIA.

"Tell me about a time you handled a difficult stakeholder."

> "At Apple, a program owner wanted the dashboard to show completion rates that excluded certain learner cohorts without documentation. I pushed back --- not as a refusal, but as a governance question. I asked: what would we tell the auditor if the denominator changed without a

> written rationale? We documented the business rule, got sign-off from the L&D director, and built an optional filter with a visible label. The program owner thanked me later when a compliance review came up."

"Where do you want to be in 3 years?"

> "I want to be leading an analytics function that is genuinely AI-augmented --- where the team spends the majority of their time on insight and decision support, not data wrangling. I am building toward that by deepening my Snowflake Cortex knowledge and exploring agentic workflows. I want the team I lead to be a competitive advantage for the business, not just a reporting function."

"Why are you leaving your current role?"

> "I have built something strong at Apple through Wipro --- the dashboards are in production, the governance is solid, and the team is capable. I am ready for a role where I own the analytics function rather than contributing to a client engagement. I want to set the data strategy, not just deliver against someone else's."

Day 7: Final Preparation and Live Simulation

Morning: Consolidate

Do not learn anything new today. Today is about confidence, not content.

1. Read your 5 STAR stories out loud. Time yourself. Each should be 90 to 120 seconds.
2. Answer the verbal pitch out loud three times without looking at notes.
3. Run the interview quiz in Test Mode. Aim for 70%+.

   Afternoon: Logistics

   - Confirm the interview format: is there a live coding round?
   - If live coding: prepare your Snowflake SQL environment or ask what tool they use
   - Test your video setup, background, and microphone
   - Have your portfolio open in a browser tab ready to share: sri-vaishnavi-1988.github.io

   Questions to Ask the Interviewer

   Always prepare questions. These signal seniority:

1. "What does the analytics team's relationship with data engineering look like? Who owns the pipeline quality?"
2. "How are KPI definitions managed today? Is there a data dictionary or governance process?"
3. "What does success look like in this role after 90 days?"
4. "Is the team using or exploring AI-assisted analytics tooling? What is the leadership's stance on it?"
5. "What is the biggest data quality challenge the team is dealing with right now?"

The Day-Of Ritual

1. Eat something. Do not interview hungry.
2. Open your portfolio and the quiz an hour before --- not to study, but to remind yourself what you have built.
3. Ten minutes before: read your verbal pitch once. Then put it away.
4. Go in as the person who built all of this --- because you did.

Part 3: Technical Reference

SQL Quick Reference

| Concept | When to use | |---|---| | `RANK()` | Ranking with gaps after ties | | `DENSE_RANK()` | Ranking without gaps after ties | | `ROW_NUMBER()` | Unique sequential number regardless of ties | | `LAG()` | Value from the previous row | | `LEAD()` | Value from the next row | | `QUALIFY` | Filter window function results without a subquery | | `CLONE` | Zero-copy table copy for testing | | `AT (OFFSET)` | Time Travel to query historical state | | CTE | Named temporary result set for a single query | | `NULLIF(x, 0)` | Prevent division by zero |

Tableau Quick Reference

| Concept | When to use | |---|---| | `FIXED` LOD | Compute at a fixed dimension, ignore view filters | | `INCLUDE` LOD | Add sub-level detail to the current view level | | `EXCLUDE` LOD | Remove a dimension for percent-of-total | | Context Filter | Run before all other filters; defines the data pool | | Table Calculation | Runs in Tableau after data is retrieved; not filterable like dimensions | |

Parameter | User-controlled input that drives calculations and filters |

Python Quick Reference

| Operation | Code pattern | |---|---| | Read CSV with dates | `pd.read_csv(f, parse_dates=["col"])` | | Null check | `df.isnull().sum()` | | Drop duplicates | `df.drop_duplicates(subset=["id"], keep="last")` | | GroupBy aggregate | `df.groupby("col").agg({"val": "mean"})` | | Conditional column | `df["flag"] = df["col"].apply(lambda x: ...)` | | Filter rows | `df[df["status"] == "completed"]` | | Large file | `pd.read_csv(f, chunksize=100_000)` | | Outlier detection | IQR: flag below Q1 - 1.5*IQR or above Q3 + 1.5*IQR |

Part 4: Salary and Negotiation

Market Rates (India, 2026)

Role	Comp Range
Analytics Manager	25--45 LPA
Senior Analytics Lead	30--55 LPA
AI-Augmented Analytics Specialist	35--60 LPA
Data Strategy / BI Architect	40--70 LPA

The Negotiation Stance

When asked for your expected salary:

> "Based on the scope of this role and my 12 years of experience including enterprise clients like NVIDIA and Apple, I am targeting the 35 to 45 LPA range. I am open to discussing the full package including variables and learning budget."

Never give a single number. Always give a range with the low end above your actual minimum.

If they push back:

> "I am flexible if the role scope and growth path align. Can you share the band for this position so we can see if there is an overlap?"

Part 5: Your Online Presence

Links to Share in Every Application

| Asset | Link | |---|---| | Portfolio | sri-vaishnavi-1988.github.io | | Skills Showcase | sri-vaishnavi-1988.github.io/skills.html | | Interview Quiz | sri-vaishnavi-1988.github.io/quiz-app/ | | LinkedIn | linkedin.com/in/srivaishnavi-devarashetty |

The Email Signature to Use Now

``` Sri Vaishnavi Devarashetty Senior Analytics Leader · Tableau · Snowflake · SQL · AI-Augmented Analytics Portfolio: sri-vaishnavi-1988.github.io LinkedIn: linkedin.com/in/srivaishnavi-devarashetty ```
```

Part 6: The Governance Rule

This one sentence, said clearly in an interview, separates you from most candidates:

> "AI drafts. I decide. Every number, query, and stakeholder message gets a human review before it leaves my hands."

Say it naturally. Mean it. It signals analytical rigour, leadership maturity, and the kind of reliability that enterprises need in a senior analytics hire.

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